RPC with GRPC and Protocol Buffers

The simple database server and its client of the previous exercise shall now communicate via gRPC [1] as an extension to our attempts building our own remote procedure calls on top of Google Protocol Buffers.

Please familiarise yourself with the way how to describe service interfaces with gRPC based on the online documentation at `grpc.io`.

Again we would like to implement the simple database server of the previous exercise.

The protocol buffer database server should support three operations:

`getRecord` reads a record of type `string` from the data base using an index of type `int32`

`addRecord` writes a record to the data base using a given index and

`getSize` determines how many records are stored in the database.

Please specify the simple database server's interface using gRPC.

Download the current version of gRPC (and possibly it corresponding Protocol Buffer version).

Choose a language of your choice from Go, C++, Java, Python, and other supported languages.

Implement the simple data base server and a corresponding client - possibly by modifying your existing implementation of server and client - in the language(s) of your choice. Note that server and client not necessarily need to be implemented in the same language. Client and server are supposed to run in different processes as separate programs.

For testing the socket client shall again write the records of the following table to the data base:

4101	Appen
4102	Ahrensburg
4103	Wedel
4104	Aumühle
4105	Seevetal
4106	Quickborn

and read the records with index 4103 and 4107.

Please also invoke the operation `getSize()` in order to find out if the correct number of records is present in the database.

Please be prepared for face-to-face technical discussions. These will happen during the on-site tutorial sessions.

Links

[1] `https://grpc.io/`